SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I AB. SAI SARANYA (1921260020) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Saranya.ab.

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	1	2	2
Method Selection	20%	2	2	2	1	1
Solution Process	25%	2	2	2	2	2
Accuracy of Calculation	20%	2	1	2	1	1
Interpretation of Results	15%	1	2	1	2	1

Total Marks 37 /40

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each

department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic	Mostly correct steps	Disorganized steps; multiple	No clear process

		steps with correct approach throughout	with minor mistakes	errors	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ANSWERS TO ANALYTICAL PROBLEM SOLVING

Course: Computer Networks (CSA07) | CO2 | Assessment Tool 1

Name: **AB Sai Saranya** | Register Number: **1921260020**

Question 1

A university network is assigned the IP address block 192.168.20.0/24. It is divided into subnets for three labs using fixed masks: Lab A /26, Lab B /27 and Lab C /28. Determine the network address, valid host range and broadcast address of each subnet, and the number of usable hosts in each.

1. Understanding the problem

The parent block 192.168.20.0/24 has 32 – 24 = 8 host bits, i.e. 2^8 = 256 addresses (192.168.20.0 to 192.168.20.255). Each lab must receive a block whose size is fixed by its prefix length. Only the last octet changes, so every calculation is done on the 4th octet.

2. Method

- Host bits = 32 – prefix; block size = $2^{\text{host bits}}$.
- Network address = first address of the block; broadcast address = last address of the block (all host bits = 1).
- Valid hosts = network + 1 to broadcast – 1; usable hosts = 2^n – 2.
- Blocks are allocated one after another without overlap, starting from the largest block (/26) so that every block starts on a correct boundary (a multiple of its block size).

3. Subnet mask and block size

Lab	Prefix	Subnet mask	Host bits	Block size	Usable hosts (2^n–2)
A	/26	255.255.255.192	6	64	64 – 2 = 62
B	/27	255.255.255.224	5	32	32 – 2 = 30
C	/28	255.255.255.240	4	16	16 – 2 = 14

4. Allocation (step by step)

- **Lab A (/26):** starts at 192.168.20.0. Block of 64 covers .0 to .63.
- **Lab B (/27):** starts at the next free address, .64 (a multiple of 32). Block of 32 covers .64 to .95.
- **Lab C (/28):** starts at .96 (a multiple of 16). Block of 16 covers .96 to .111.

5. Result

Lab	Network address	Valid host range	Broadcast address	Usable hosts
A (/26)	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
B (/27)	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
C (/28)	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

6. Verification and interpretation

- Binary check for Lab B: the 4th octet 64 = **010**00000; the first three bits (010) are the subnet bits and the last five are host bits. All host bits 0 gives .64 (network), all host bits 1 gives 01011111 = .95 (broadcast).
- The three blocks do not overlap: .0–.63, .64–.95, .96–.111.
- Addresses used: 64 + 32 + 16 = 112 of 256. The remaining range 192.168.20.112 – 192.168.20.255 (144 addresses) is free for future labs.

Conclusion: Lab A can host 62 devices, Lab B 30 devices and Lab C 14 devices. Choosing the mask according to the number of devices in each lab avoids wasting addresses (VLSM approach).

Question 2

System 1 has IP address 192.168.1.130 and System 2 has IP address 192.168.1.190, both with subnet mask 255.255.255.192. Determine the subnet range, network address and broadcast address for each system, the valid host range, whether each IP falls within its subnet, and the number of usable hosts.

1. Understanding the problem

The mask 255.255.255.192 is equivalent to /26 (255.255.255.**11000000**). Two bits of the 4th octet are used for the subnet and six bits remain for hosts. We must find the subnet in which each address lies by using the mask.

2. Method

- Block size = 256 – 192 = 64, so the subnets in 192.168.1.0 are .0, .64, .128 and .192 (2^2 = 4 subnets).
- Network address = IP address **AND** subnet mask.
- Broadcast address = network address + block size – 1 (host bits all 1).
- Usable hosts = $2^6 - 2 = 62$.

3. Subnet table for 192.168.1.0 with mask 255.255.255.192

Subnet	Network	Host range	Broadcast
1	192.168.1.0	.1 – .62	192.168.1.63
2	192.168.1.64	.65 – .126	192.168.1.127
3	192.168.1.128	.129 – .190	192.168.1.191
4	192.168.1.192	.193 – .254	192.168.1.255

4. Calculation using the bitwise AND

System 1 (192.168.1.130): 4th octet 130 = 10000010; mask 192 = 11000000; AND = 10000000 = 128. Network address = **192.168.1.128**.

System 2 (192.168.1.190): 4th octet 190 = 10111110; mask 192 = 11000000; AND = 10000000 = 128. Network address = **192.168.1.128**.

Both systems produce the same network address, so they are in the **same subnet**.

5. Result

Parameter	System 1 (192.168.1.130)	System 2 (192.168.1.190)
Subnet mask	255.255.255.192 (/26)	255.255.255.192 (/26)
Network address	192.168.1.128	192.168.1.128
Subnet range	192.168.1.128 – 192.168.1.191	192.168.1.128 – 192.168.1.191
Valid host range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190
Broadcast address	192.168.1.191	192.168.1.191
Usable hosts	62	62
IP within host range?	Yes (130 lies between 129 and 190)	Yes (190 is the last valid host address)

6. Interpretation

- Both IP addresses fall inside the valid host range 129–190 of the subnet 192.168.1.128/26. Neither is the network address (.128) or the broadcast address (.191), so both are legitimate host addresses.
- Note that 192.168.1.190 is the *highest* usable address of the subnet. Had it been .191 it would be the broadcast address and invalid for a host.
- Because the systems share the same network ID, they can communicate directly (layer 2 switching) without a router. The subnet offers 62 usable host addresses, from .129 to .190.

Conclusion: Systems 1 and 2 belong to the single subnet 192.168.1.128/26 (range .128–.191, broadcast .191) with 62 usable host addresses.

Question 3

An organization has the block 172.16.0.0/16. Department masks are fixed: HR /25, Sales /26, IT /27, Admin /28. Assign subnet addresses, find the network address, valid host range and broadcast address of each subnet, the usable hosts per department, and the remaining unused address range.

1. Understanding the problem

The block 172.16.0.0/16 has 16 host bits (65,536 addresses: 172.16.0.0 – 172.16.255.255). The departments need blocks of different sizes, so Variable Length Subnet Masking (VLSM) is used. The allocation proceeds from the largest block to the smallest so that each block begins at a valid boundary and no space is wasted by misalignment.

2. Block sizes

Department	Prefix	Mask	Host bits	Block size	Usable hosts
HR	/25	255.255.255.128	7	128	126
Sales	/26	255.255.255.192	6	64	62
IT	/27	255.255.255.224	5	32	30
Admin	/28	255.255.255.240	4	16	14

3. Sequential allocation

- **HR /25:** start at 172.16.0.0; 128 addresses cover .0 – .127.
- **Sales /26:** next free address is 172.16.0.128 (a multiple of 64); 64 addresses cover .128 – .191.
- **IT /27:** next free is 172.16.0.192 (a multiple of 32); 32 addresses cover .192 – .223.
- **Admin /28:** next free is 172.16.0.224 (a multiple of 16); 16 addresses cover .224 – .239.

4. Result

Dept.	Network address	Valid host range	Broadcast	Usable hosts
HR /25	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales /26	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT /27	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin /28	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

5. Remaining unused address range

The last address allocated is 172.16.0.239. Therefore the unused portion of the /16 block is:

172.16.0.240 – 172.16.255.255

- 172.16.0.240 – 172.16.0.255 = 16 addresses (the remainder of the first /24).
- 172.16.1.0 – 172.16.255.255 = 255 × 256 = 65,280 addresses.
- Total used = 128 + 64 + 32 + 16 = 240 addresses; total unused = 65,536 – 240 = **65,296** addresses (16 + 65,280 = 65,296, which confirms the result).

6. Interpretation

The four departments together occupy only 240 addresses (about 0.37% of the /16), all inside 172.16.0.0/24. The large unused space (e.g. 172.16.1.0 onwards) leaves ample room for expansion, new departments, servers or WAN links. Allocating the biggest block first keeps every subnet aligned and contiguous, which also allows future route summarization. The 2 addresses per subnet reserved for network and broadcast give 126, 62, 30 and 14 usable hosts respectively.

Question 4

A company uses 172.16.10.0/24 and subnets it with a /27 mask. Calculate the number of subnets and usable hosts per subnet, find the subnet of 172.16.10.78 with its network and broadcast addresses, and decide whether 172.16.10.95 falls in the same subnet.

1. Understanding the problem

The original network is a /24 (24 network bits, 8 host bits). The /27 mask (255.255.255.224) borrows 27 – 24 = **3 bits** from the host portion, leaving 5 host bits.

2. Formulae and calculation

- Number of subnets = $2^{\text{borrowed bits}} = 2^3 =$ **8**.
- Hosts per subnet = $2^5 - 2 = 32 - 2 =$ **30**.
- Block size = 256 – 224 = 32 (the subnets increase in steps of 32).

3. List of all eight subnets

No.	Network address	Host range	Broadcast
1	172.16.10.0	.1 – .30	172.16.10.31
2	172.16.10.32	.33 – .62	172.16.10.63
3	172.16.10.64	.65 – .94	172.16.10.95
4	172.16.10.96	.97 – .126	172.16.10.127
5	172.16.10.128	.129 – .158	172.16.10.159
6	172.16.10.160	.161 – .190	172.16.10.191
7	172.16.10.192	.193 – .222	172.16.10.223
8	172.16.10.224	.225 – .254	172.16.10.255

4. Subnet of 172.16.10.78

Using the AND operation: 78 = 01001110; mask 224 = 11100000; AND = 01000000 = 64. Alternatively, 78 lies between 64 and 95 (64 + 32 = 96), so it belongs to subnet 3.

- Network address: **172.16.10.64**
- Valid host range: 172.16.10.65 – 172.16.10.94
- Broadcast address: **172.16.10.95** (64 + 32 – 1; binary 010**11111**)

5. Is 172.16.10.95 in the same subnet?

95 = 01011111; AND with 224 (11100000) = 01000000 = 64, so 172.16.10.95 gives the **same network address 172.16.10.64**. It therefore lies within the same subnet block (172.16.10.64 – 172.16.10.95) as 172.16.10.78.

However, 172.16.10.95 is the **broadcast address** of that subnet (all host bits = 1). It is part of the subnet range but is **not a valid host address** and cannot be assigned to a device; packets sent to it are delivered to every host in the subnet 172.16.10.64/27.

6. Interpretation

Subnetting with /27 splits the /24 into eight equal subnets of 30 usable hosts each, suitable for departments of up to 30 devices. The host 172.16.10.78 sits inside subnet 3, and the address .95 marks the end of that subnet; the next address, 172.16.10.96, belongs to a different subnet (subnet 4) and would need a router to communicate with .78.

Conclusion: 8 subnets, 30 usable hosts each; 172.16.10.78 is in 172.16.10.64/27 (broadcast 172.16.10.95); 172.16.10.95 is in the same subnet range but only as its broadcast address.

Question 5

Given the IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329: compress it to its shortest form, expand the compressed form back to full notation, find the network prefix for a /64 prefix length, and calculate the total number of host addresses in the network.

1. Understanding the problem

An IPv6 address is 128 bits long, written as eight groups of four hexadecimal digits separated by colons (each group = 16 bits). Two rules allow shortening: (i) leading zeros in any group may be omitted; (ii) one single sequence of consecutive all-zero groups may be replaced by "::" (only once in an address, to avoid ambiguity).

2. Compression

Original address:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

Step 1 – remove leading zeros in each group:

```
2001:db8:0:0:0:ff00:42:8329
```

Step 2 – replace the longest run of consecutive zero groups (groups 3, 4 and 5) with "::":

```
2001:db8::ff00:42:8329
```

Shortest form: **2001:db8::ff00:42:8329**

3. Expansion of the compressed address

- Write the groups: 2001, db8, [::], ff00, 42, 8329. Only 5 groups are visible, so "::" stands for 8 – 5 = **3** zero groups.
- Insert three groups of 0000 at the position of "::": 2001 : db8 : 0 : 0 : 0 : ff00 : 42 : 8329.
- Pad every group with leading zeros to four digits.

Full form: **2001:0db8:0000:0000:0000:ff00:0042:8329**, which is identical to the original address, confirming the compression is correct.

4. Network prefix with /64

A /64 prefix means the first 64 bits (the first four groups) identify the network and the last 64 bits (the next four groups) form the interface identifier (host part).

Part	Groups	Value
Network prefix (64 bits)	1 – 4	2001:0db8:0000:0000
Interface ID (64 bits)	5 – 8	0000:ff00:0042:8329

Network prefix: **2001:db8:0:0::/64**, written in shortest form as **2001:db8::/64** (full form 2001:0db8:0000:0000:0000:0000:0000:0000/64, obtained by setting all host bits to 0). The interface identifier of the given host is ::ff00:42:8329.

5. Number of host addresses

Host bits = 128 – 64 = 64. Number of addresses = 2^{64} = **18,446,744,073,709,551,616** (about 1.8×10^{19}) addresses.

IPv6 has no broadcast address (it uses multicast instead), so there is no –2 deduction for a broadcast. By convention the all-zero interface ID is reserved as the Subnet-Router anycast address, but the size of the address space of the network is 2^{64}.

6. Interpretation

A /64 is the standard subnet size in IPv6 (required for SLAAC auto-configuration), so every LAN receives an enormous address space; a single /64 is about four billion times larger than the entire IPv4 Internet (2^{32} addresses). Address conservation is therefore not a concern in IPv6, and subnet design focuses on hierarchy and ease of management instead.

Conclusion: Compressed: 2001:db8::ff00:42:8329; expanded: 2001:0db8:0000:0000:0000:ff00:0042:8329; network prefix: 2001:db8::/64; total addresses: 2^{64} = 18,446,744,073,709,551,616.